# UNIT - IV : FINANCIAL MANAGEMENT

## I. Scope and nature of financial management

* FINANCIAL MANAGEMENT = planning, organising, directing and controlling the financial activities of an enterprise - procurement and effective utilisation of funds. (Solomon: the efficient use of an important economic resource, namely capital funds.) Objective: PROFIT maximisation (short term) and WEALTH maximisation (long term - maximising the market value of the shareholders' equity) with proper risk balance.

## Scope - the three decisions




* Nature: it is a part of overall management; continuous; deals with money (cash flows), risk and return trade-off; both an ART and a SCIENCE; provides inputs for all functional areas; involves forecasting, planning, control, and reporting; applies to all types of business.
* Functions of the finance manager: estimating fund requirements, raising funds at least cost, capital structure decision, investment (fixed assets and working capital) decisions, dividend policy, cash

management, financial control and analysis (ratios), tax planning, protecting the funds (insurance, risk).

## 2. Sources of Finance

| Basis | Sources |
|---|---|
| LONG TERM (> 5 years) | Equity (ordinary) shares, preference shares, debentures / bonds, long-term loans from banks / financial institutions (IDBI, SIDBI, IFCI), retained earnings (ploughing back), venture capital, lease financing, public deposits (3 yr) |
| MEDIUM TERM (1 - 5 yr) | Term loans, lease and hire purchase, public deposits, medium term debentures, commercial paper, loan from commercial banks |
| SHORT TERM (< 1 yr) | Trade credit, bank overdraft, cash credit, bills discounting, factoring, commercial paper, advances from customers, accrued expenses, short term loans |

| Internal sources | External sources |
|---|---|
| Retained earnings, depreciation funds, reserves, sale of assets | Shares, debentures, loans, deposits, trade credit, lease, venture capital, IPO |

### Features of important sources

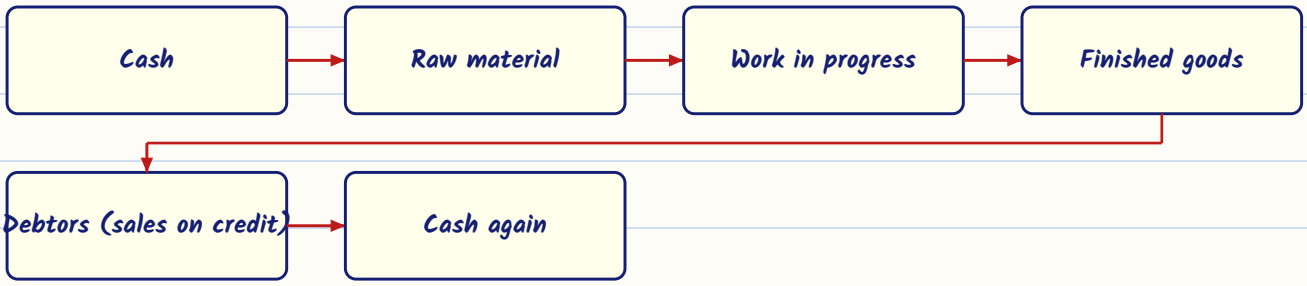
* EQUITY SHARES: owners' capital, ownership and voting rights, no fixed dividend (residual claim), no repayment (permanent), high risk and cost. PREFERENCE SHARES: fixed dividend, preference over equity in dividend and repayment, usually no voting rights; cumulative / non-cumulative, redeemable / irredeemable, convertible.
* DEBENTURES: loan capital, fixed interest, repayable on maturity,

interest is tax-deductible, no ownership dilution; secured / unsecured, convertible / non-convertible. RETAINED EARNINGS: cheapest internal source, no flotation cost. TERM LOANS: from banks and financial institutions, fixed repayment schedule. LEASE: use of asset without buying (operating and finance lease). VENTURE CAPITAL: funds for risky start-ups in exchange for equity.

| Equity | Debt |
|---|---|
| Ownership, voting | No ownership |
| Dividend not mandatory | Interest is compulsory |
| No repayment | Repayable |
| Dividend NOT tax deductible | Interest IS tax deductible |
| Higher cost of capital | Lower cost, but financial risk |

## 3. Management of Working Capital

* WORKING CAPITAL = capital needed for day to day operations. GROSS working capital = total current assets. NET working capital = current assets - current liabilities. Current assets: cash, bank, marketable securities, debtors (receivables), bills receivable, stock (raw materials, WIP, finished goods), prepaid expenses. Current liabilities: creditors, bills payable, outstanding expenses, short term loans, provisions.
* Types: permanent (regular / fixed) working capital - minimum level always needed, and temporary (variable / seasonal) working capital - fluctuates with seasons and demand.




The OPERATING CYCLE (working capital cycle) = time from cash paid for raw material to cash collected from customers:

**Operating cycle = RM holding period + WIP period + FG holding period + D**

**Cash cycle = Operating cycle - Creditors payment period**

## Importance and management

* Too little working capital -> stock-outs, inability to pay creditors, loss of goodwill. Too much -> idle funds, low return, risk of theft / obsolescence. The manager balances PROFITABILITY and LIQUIDITY.
* Management techniques: CASH management (cash budget, Baumol model, float); RECEIVABLES management (credit policy, credit period, discounts, ageing schedule); INVENTORY management (EOQ, ABC analysis, JIT, safety stock, reorder level); PAYABLES management (credit terms); short term financing (bank credit, commercial paper). Current ratio = CA / CL (ideal 2:1); Quick ratio = (CA - stock)/CL (ideal 1:1).

## Estimation of working capital requirements

* Method: estimate the investment in each current asset for the projected activity level, subtract current liabilities, and add a

CONTINGENCY margin (10 - 20%).

**Working capital = Current assets (stock, debtors, cash, WIP) - Current li**

Worked example: Annual sales (all on credit) = 12,00,000; cost of production = 10,00,000 of which raw material 4,00,000, labour 3,00,000, overheads 3,00,000. Raw material stock 1 month, WIP 0.5 month, finished goods 1 month, debtors 1.5 months, creditors 1 month, wages / overheads paid 1 month in arrears, cash balance 40,000, contingency 10%.

| Item | Calculation | Amount (Rs) |
|---|---|---|
| Raw material stock | 4,00,000 x 1/12 | 33,333 |
| Work in progress | 10,00,000 x 0.5/12 | 41,667 |
| Finished goods | 10,00,000 x 1/12 | 83,333 |
| Debtors (at cost) | 10,00,000 x 1.5/12 | 1,25,000 |
| Cash balance | given | 40,000 |
| Total current assets | | 3,23,333 |
| Creditors | 4,00,000 x 1/12 | 33,333 |
| Outstanding wages and overheads | 6,00,000 x 1/12 | 50,000 |
| Total current liabilities | | 83,333 |
| Net working capital | 3,23,333 - 83,333 | 2,40,000 |
| Add 10% contingency | | 24,000 |
| WORKING CAPITAL REQUIRED | | 2,64,000 |

## 4. Budget and Budgetary Control

* BUDGET = a detailed plan of operations for a future period expressed

in quantitative / monetary terms (CIMA: a financial and / or quantitative statement prepared prior to a defined period of time of the policy to be pursued to attain a given objective). BUDGETARY CONTROL = establishing budgets, continuously comparing ACTUAL with BUDGETED results, finding VARIANCES and taking corrective action.

## Objectives and steps

* Planning and coordination, control of costs, communication, motivation, performance evaluation, fixing responsibility, optimum use of resources. Steps: set organisation objectives -> budget committee and budget manager -> identify the LIMITING (principal budget) factor (usually sales) -> prepare budgets -> coordinate and approve -> implement -> compare actual with budget -> analyse variances -> corrective action.

## Types of budgets

| Type | Examples / meaning |
|---|---|
| Functional budgets | Sales, production, raw material purchase, labour, overhead, cost of goods sold, capital expenditure, research budget |
| Master budget | Summary of all functional budgets: budgeted P&L account and balance sheet |
| Fixed budget | Prepared for ONE level of activity, not changed |
| Flexible budget | Adjusted for different activity levels (variable cost changes with output) |
| Cash budget | Estimated cash receipts and payments |
| Zero base budget (ZBB) | Every item justified afresh from zero, ignoring past |
| Performance / programme budget | Based on programmes and expected results |

* Advantages: discipline in planning, coordination, cost consciousness, early warning, goal congruence. Limitations: based on estimates, rigid if not flexible, time consuming and costly, resistance from managers, cannot replace management.

## 5. Capital Budgeting

* CAPITAL BUDGETING = planning, evaluating and selecting long term investment proposals (acquiring fixed assets - machinery, new plant, new product, expansion, replacement) whose benefits are expected over a number of years. Also called capital expenditure decision / investment appraisal.

## Nature of investment decisions

* Large funds committed; LONG term effect; IRREVERSIBLE (high cost of reversal); strategic - affects growth, profitability and risk of the firm; involves uncertainty about future cash flows; types: expansion, replacement and modernisation, diversification, independent / mutually exclusive / contingent projects; decisions need accurate cash flow forecasting (after tax, incremental, including depreciation tax shield and working capital).

## Investment evaluation criteria

| Non-discounting (traditional) | Discounting (time value of money) |
|---|---|
| Payback period (PBP) | Net Present Value (NPV) |
| Accounting rate of return (ARR) | Internal Rate of Return (IRR) |
| | Profitability Index (PI) / Benefit - cost ratio |

## (1) Payback period (PBP)

* Time required to recover the initial investment from net cash inflows. Accept if PBP < standard period. Merits: simple, favours liquidity, risk indicator. Demerits: ignores time value of money and cash flows after payback; ignores profitability.

**PBP (even cash flows) = Initial investment / Annual cash inflow**

**PBP (uneven) = Years before full recovery + (Unrecovered amount / Cash**

## (2) Accounting (Average) Rate of Return (ARR)

**ARR = (Average annual profit after tax / Average (or initial) investment)**

* Average investment = (Initial investment + Salvage value) / 2. Accept

if ARR > required rate. Uses accounting profit, not cash flow; ignores time value.

## (3) Net Present Value (NPV)

$$NPV = \text{sum from } t = 1 \text{ to } n \text{ of } C_t / (1 + k)^t - I_0$$

* $C_t$ = cash inflow in year t, k = discount rate (cost of capital), $I_0$ = initial investment. DECISION: accept if NPV > 0; reject if NPV < 0; among mutually exclusive projects choose the highest NPV. Merits: time value, whole life, maximises wealth. Demerit: needs cost of capital; absolute measure.

## (4) Internal Rate of Return (IRR)

* The discount rate at which NPV = 0 (present value of inflows = outflow). Found by trial and error / interpolation:

$$IRR = r_1 + [\ NPV_1 / (NPV_1 - NPV_2)\ ] \times (r_2 - r_1)$$

* Accept if IRR > cost of capital (k). Merits: percentage, time value. Demerits: multiple IRRs for non-conventional flows; may conflict with NPV for mutually exclusive projects (assumes reinvestment at IRR).

## (5) Profitability Index (PI)

$$PI = \text{Present value of cash inflows} / \text{Initial investment} = 1 + NPV / I_0$$

* Accept if PI > 1. Useful for ranking projects under capital rationing.

## Numerical problem

A project needs an investment of Rs 1,00,000 and gives net cash inflows of Rs 40,000, 40,000, 40,000 and 30,000 in years 1 - 4. Cost of capital = 10%. Scrap value nil. Evaluate by all methods.

| Year | Cash flow | PV factor 10% | PV (Rs) | Cumulative CF |
|---|---|---|---|---|
| 0 | -1,00,000 | 1.000 | -1,00,000 | -1,00,000 |
| 1 | 40,000 | 0.909 | 36,360 | -60,000 |
| 2 | 40,000 | 0.826 | 33,040 | -20,000 |
| 3 | 40,000 | 0.751 | 30,040 | +20,000 |
| 4 | 30,000 | 0.683 | 20,490 | +50,000 |

* PBP: recovered during year 3: 2 + 20,000/40,000 = 2.5 years.
* NPV = (36,360 + 33,040 + 30,040 + 20,490) - 1,00,000 = 1,19,930 - 1,00,000 = + Rs 19,930 -> ACCEPT.
* PI = 1,19,930 / 1,00,000 = 1.20 (> 1) -> accept.
* ARR: total inflow = 1,50,000, net profit over life (before depreciation treatment) = 50,000 -> average annual profit = 12,500 -> ARR on initial investment = 12,500/1,00,000 = 12.5%; on average investment (50,000) = 25%.
* IRR: try 18%: PV = 40,000 (0.847 + 0.718 + 0.609) + 30,000 (0.516) = 86,960 + 15,480 = 1,02,440 -> NPV = +2,440. Try 20%: 40,000 (0.833 + 0.694 + 0.579) + 30,000 (0.482) = 84,240 + 14,460 = 98,700 -> NPV = -1,300.

**IRR = 18 + [2,440 / (2,440 + 1,300)] x 2 = 18 + 1.30 = 19.3% (> 10%, ac**

## Second example - choosing between two projects

| | Project A | Project B |
|---|---|---|
| Investment | Rs 50,000 | Rs 50,000 |
| Cash inflows yr 1 to 3 | 25,000 each | 10,000, 25,000, 40,000 |
| PBP | 2.0 years | 2.4 years |
| NPV at 10% | 25,000 x 2.487 - 50,000 = +12,175 | 9,090 + 20,650 + 30,040 - 50,000 = +9,780 |
| PI | 1.24 | 1.20 |
| Decision | A is preferred (higher NPV, faster payback) | |

| Criterion | Accept if | Time value? |
|---|---|---|
| PBP | PBP < standard | No |
| ARR | ARR > required rate | No |
| NPV | NPV > 0 | Yes |
| IRR | IRR > cost of capital | Yes |
| PI | PI > 1 | Yes |

## Time value of money (basics)

**Future value FV = P (1 + r)$^n$ ; Present value PV = F / (1 + r)$^n$ ; PV of ann**

* A rupee today is worth more than a rupee tomorrow (inflation, opportunity cost, risk). Compounding moves money forward in time; discounting brings future cash flows to the present.

Revision: 3 finance decisions - investment, financing, dividend. WC = CA - CL. Operating cycle. Budget types: fixed, flexible, ZBB. Capital budgeting: PBP, ARR (non-discounting); NPV > 0, IRR > k, PI > 1 (discounting).